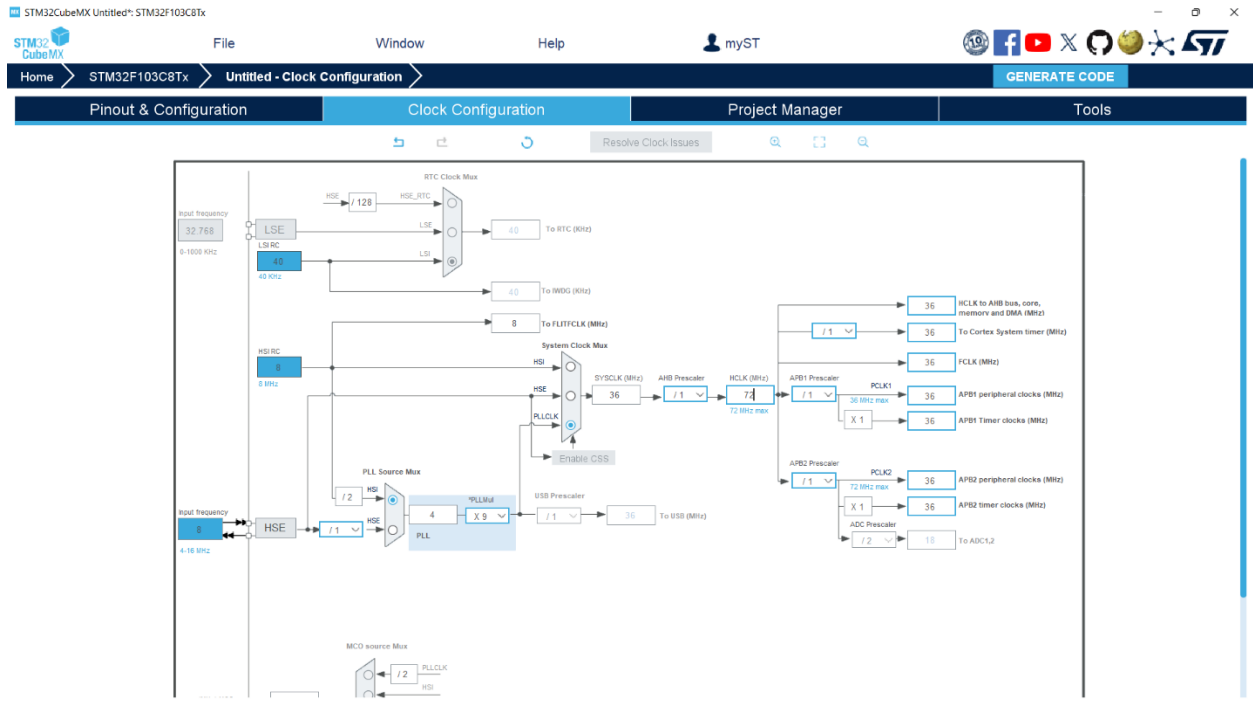


تقرير واجب (2)

ضبط تردد الساعة على 72 هرتز باستخدام البلورة الخارجية



تهيئة I2C2

STM32CubeMX Untitled: STM32F103C8Tx

File Window Help myST

Home > STM32F103C8Tx > Untitled - Pinout & Configuration > GENERATE CODE

Pinout & Configuration Clock Configuration Project Manager Tools

Software Packs Pinout

I2C2 Mode and Configuration

I2C2 Mode

I2C2

Configuration

Reset Configuration

NVIC Settings DMA Settings GPIO Settings

Parameter Settings User Constants

Search Signals

Search (Ctrl+F)

Pin Name Signal on Pin GPIO output I. GPIO mode GPIO Pull-up Maximum out. User L

PB10	I2C2_SCL	n/a	Alternate Fun...	n/a	High	
PB11	I2C2_SDA	n/a	Alternate Fun...	n/a	High	

Pinout view System view

Pinout view

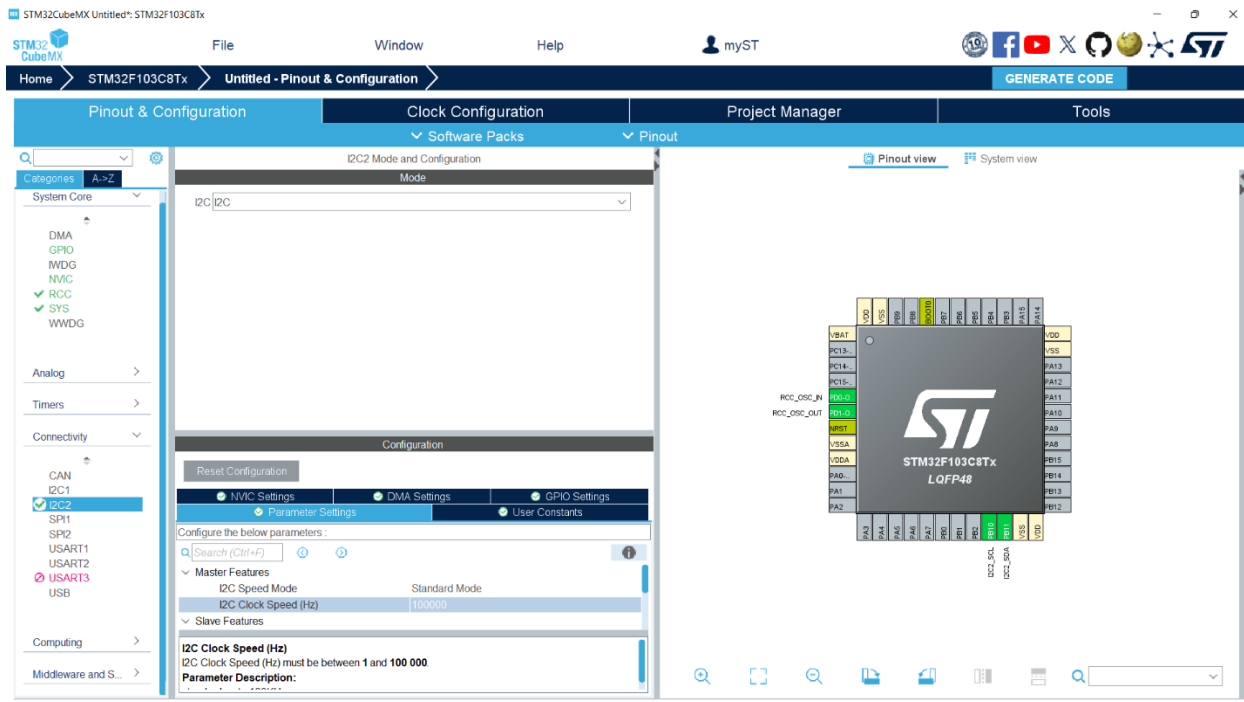
System view

STM32F103C8Tx LQFP48

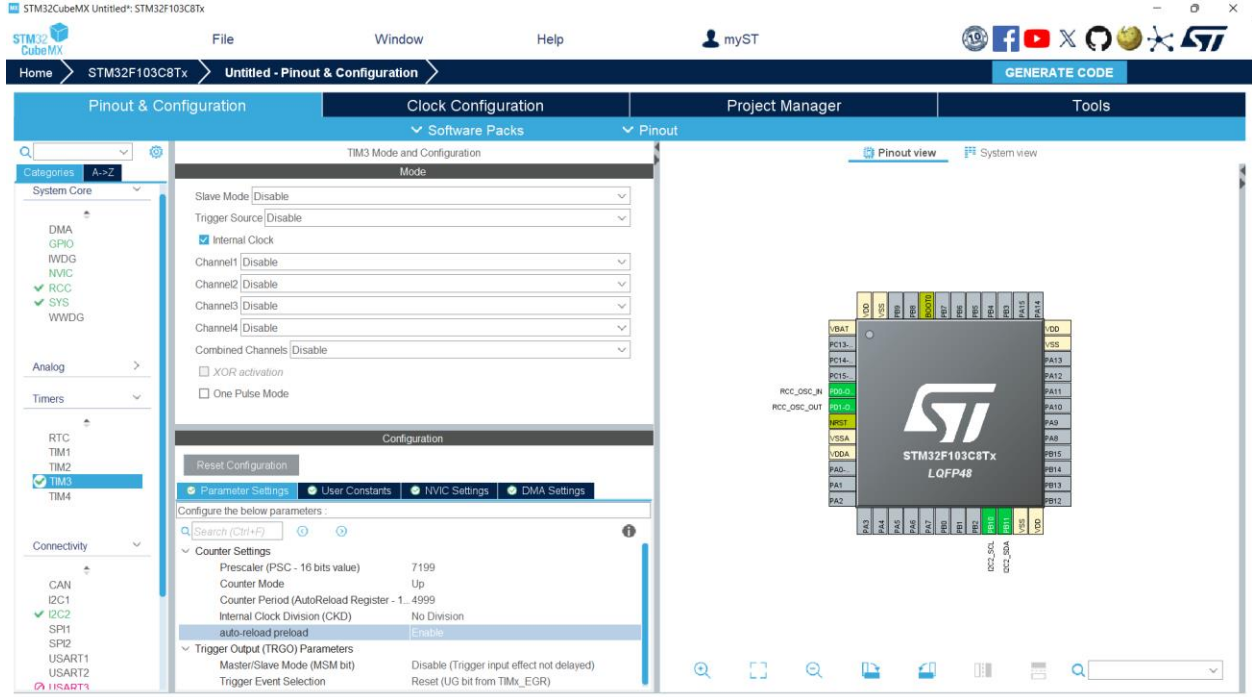
Pinout view

System view

ضبط السرعة على 100 كيلوهرتز



تهيئة المؤقت للحصول على مقاطعة كل 500 ميلي ثانية



تفعيل المقاطعة

STM32CubeMX Untitled: STM32F103C8Tx

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Home STM32F103C8Tx Untitled - Pinout & Configuration GENERATE CODE

Pinout & Configuration Clock Configuration Project Manager Tools

Software Packs Pinout

Categories A-Z

System Core

- DMA
- GPIO
- MDG
- NVIC
- ✓ RCC
- ✓ SYS
- WWDG

Analog

Timers

- RTC
- TIM1
- TIM2
- ✓ TIM3
- TIM4

Connectivity

- CAN
- ✓ I2C1
- ✓ I2C2
- SP1
- SP2
- USART1
- USART2
- ✓ USART3

TIM3 Mode and Configuration

Mode

Slave Mode Disable

Trigger Source Disable

☒ Internal Clock

Channel1 Disable

Channel2 Disable

Channel3 Disable

Channel4 Disable

Combined Channels Disable

☐ XOR activation

☐ One Pulse Mode

Configuration

Reset Configuration

Parameter Settings User Constants NVIC Settings DMA Settings

NVIC Interrupt Table

Enabled	Preemption Priority	Sub Priority
☒	0	0

TIM3 global interrupt

Pinout view System view

STM32F103C8Tx LQFP48

VBAT VSS PC13- PA13 VDD PA15 PA14 PA12 PA11 PA10 PA9 PA8 PA7 PA6 PA5 PA4 PA3 PA2 PA1 PA0 PA2 PA4 PA6 PA7 PA8 PA9 PA10 PA11 PA12 PA13 PA14 PA15 VDD VSS

RCC_OSC_IN RCC_OSC_OUT

DC2_SCL DC2_SDA

تهيئة ال GPIOs

STM32CubeMX Untitled: STM32F103C8Tx

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Home STM32F103C8Tx Untitled - Pinout & Configuration GENERATE CODE

Pinout & Configuration Clock Configuration Project Manager Tools

GPIO Mode and Configuration

Configuration

Show All

GPIO

Search Signals

Search (Ctrl+F)

Show only Modified Pins

Pin Name	Signal on Pin	GPIO output lev.	GPIO mode	GPIO Pull-up/P	Maximum outp.	User Label	Modified
PA0-WKUP	n/a	n/a	Input mode	Pull-up	n/a	Button_Manual	☒
PA1	n/a	n/a	Input mode	Pull-up	n/a	Button_Manual	☒
PA2	n/a	n/a	Input mode	Pull-up	n/a	Button_Auto	☒
PB0	n/a	Low	Output Push Pull	No pull-up and	Medium	Motor_Control	☒
PB10	I2C2_SCL	n/a	Alternate Funct.	n/a	High		☐
PB11	I2C2_SDA	n/a	Alternate Funct.	n/a	High		☐
PD0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	n/a		☐
PD1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	n/a		☐

~PA1 Configuration :

GPIO mode: Input mode

GPIO Pull-up/Pull-down: Pull-up

User Label: Button_Manual_Off

Pinout view System view

Project manager & code generator

STM32CubeMX Assignment2.ioc STM32F103C8Tx

File Window Help myST

Home STM32F103C8Tx Assignment2.ioc - Project Manager GENERATE CODE

Pinout & Configuration	Clock Configuration	Project Manager	Tools
Project	Project Settings Project Name Assignment2 Project Location C:\Users\cec20\Downloads\Telegram Desktop\stm32mm Browse Application Structure Advanced ☐ Do not generate the main() Toolchain Folder Location C:\Users\cec20\Downloads\Telegram Desktop\stm32mm\Assignment2\ Toolchain / IDE EWARM Min Version V8.32 ☐ Generate Under Root		
Code Generator			
Advanced Settings	Linker Settings Minimum Heap Size 0x200 Minimum Stack Size 0x400 Thread-safe Settings Cortex-M3NS ☐ Enable multi-threaded support Thread-safe Locking Strategy Default - Mapping suitable strategy depending on RTOS selection Mcu and Firmware Package Mcu Reference STM32F103C8Tx Firmware Package Name and Version STM32Cube_FW_F1_V1.8.7 ☒ Use latest available version ☒ Use Default Firmware Location Firmware Relative Path C:\Users\cec20\STM32Cube\Repository\STM32Cube_FW_F1_V1.8.7 Browse		

الكود البرمجي : جزء تعريف المتغيرات العامة و عنوان الحساس

```
/* USER CODE BEGIN PV */
#define TC74_ADDR 0x9A
uint8_t temperature = 0;
uint8_t auto_mode_flag = 1;
/* USER CODE END PV */
```

الكود البرمجي : جزء دالة قراءة الحرارة

```
/* USER CODE BEGIN 0 */
uint8_t TC74_ReadTemp(void) {
    uint8_t temp_byte = 0;
    if (HAL_I2C_Master_Receive(&hi2c2, TC74_ADDR, &temp_byte, 1, 100) == HAL_OK) {
        return temp_byte;
    } else {
        return 0;
    }
}
/* USER CODE END 0 */
```


الكود البرمجي :جزء تشغيل المؤقت

```
/* USER CODE BEGIN 2 */
HAL_TIM_Base_Start_IT(&htim3);
/* USER CODE END 2 */
```

الكود البرمجي :جزء معالجة الأضرار الثلاثة

```
while (1)
{
    /* USER CODE BEGIN 3 */
    if (HAL_GPIO_ReadPin(Button_Manual_On_GPIO_Port, Button_Manual_On_Pin) == GPIO_PIN_RESET) {
        HAL_Delay(50);
        if (HAL_GPIO_ReadPin(Button_Manual_On_GPIO_Port, Button_Manual_On_Pin) == GPIO_PIN_RESET) {
            auto_mode_flag = 0;
            HAL_GPIO_WritePin(Motor_Control_GPIO_Port, Motor_Control_Pin, GPIO_PIN_SET);
        }
        while (HAL_GPIO_ReadPin(Button_Manual_On_GPIO_Port, Button_Manual_On_Pin) == GPIO_PIN_RESET) {}
    }
    if (HAL_GPIO_ReadPin(Button_Manual_Off_GPIO_Port, Button_Manual_Off_Pin) == GPIO_PIN_RESET) {
        HAL_Delay(50);
        if (HAL_GPIO_ReadPin(Button_Manual_Off_GPIO_Port, Button_Manual_Off_Pin) == GPIO_PIN_RESET) {
            auto_mode_flag = 0;
            HAL_GPIO_WritePin(Motor_Control_GPIO_Port, Motor_Control_Pin, GPIO_PIN_RESET);
        }
        while (HAL_GPIO_ReadPin(Button_Manual_Off_GPIO_Port, Button_Manual_Off_Pin) == GPIO_PIN_RESET) {}
    }
    if (HAL_GPIO_ReadPin(Button_Auto_GPIO_Port, Button_Auto_Pin) == GPIO_PIN_RESET) {
        HAL_Delay(50);
        if (HAL_GPIO_ReadPin(Button_Auto_GPIO_Port, Button_Auto_Pin) == GPIO_PIN_RESET) {
            auto_mode_flag = 1;
            if (temperature > 30) HAL_GPIO_WritePin(Motor_Control_GPIO_Port, Motor_Control_Pin, GPIO_PIN_SET);
            else HAL_GPIO_WritePin(Motor_Control_GPIO_Port, Motor_Control_Pin, GPIO_PIN_RESET);
        }
        while (HAL_GPIO_ReadPin(Button_Auto_GPIO_Port, Button_Auto_Pin) == GPIO_PIN_RESET) {}
    }
    HAL_Delay(10);
    /* USER CODE END 3 */
}
```

الكود البرمجي : جزء دالة المقاطعة

```
/* USER CODE BEGIN 4 */
void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef *htim) {
    if (htim->Instance == TIM3) {
        temperature = TC74_ReadTemp();
        if (auto_mode_flag == 1) {
            if (temperature > 30) HAL_GPIO_WritePin(Motor_Control_GPIO_Port, Motor_Control_Pin, GPIO_PIN_SET);
            else HAL_GPIO_WritePin(Motor_Control_GPIO_Port, Motor_Control_Pin, GPIO_PIN_RESET);
        }
    }
}
/* USER CODE END 4 */
```